

USING STEPANOV'S METHOD FOR EXPONENTIAL SUMS INVOLVING RATIONAL FUNCTIONS

TODD COCHRANE AND CHRISTOPHER PINNER

ABSTRACT. For a non-trivial additive character ψ and multiplicative character χ on a finite field $\mathbb{F}_q$, and rational functions f, g in $\mathbb{F}_q(x)$, we show that the elementary Stepanov-Schmidt method can be used to obtain the corresponding Weil bound for the sum $\sum_{x \in \mathbb{F}_q \setminus \mathcal{S}} \chi(g(x))\psi(f(x))$ where $\mathcal{S}$ is the set of the poles of f and g . We also determine precisely the number of characteristic values ω_i of modulus $q^{1/2}$ and the number of modulus 1.

1. INTRODUCTION

Let $\psi(x) = e^{2\pi i \text{tr}_{\mathbb{F}_q/\mathbb{F}_p}(b_\psi x)}$ be a non-trivial additive character and χ a multiplicative character on a finite field $\mathbb{F}_q$ of characteristic p , f, g rational functions in $\mathbb{F}_q(x)$ and

$$S(\psi, f; \chi, g) = \sum_{x \in \mathbb{F}_q \setminus \mathcal{S}} \chi(g(x))\psi(f(x)),$$

where $\mathcal{S}$ denotes the set of poles of f and g . For $f = f_1/f_2$ we define $\deg(f) = \deg(f_1) - \deg(f_2)$. The sum is said to be degenerate if both

$$(1.1) \quad b_\psi f(x) = h(x)^p - h(x) + c, \quad \text{for some } h(x) \in \mathbb{F}_q(x), c \in \mathbb{F}_q,$$

and

$$(1.2) \quad g(x) = ch(x)^n, \quad \text{for some } h(x) \in \mathbb{F}_q(x), c \in \mathbb{F}_q,$$

where n is the order of χ . For such f and g the summand is constant and so one can do no better than a trivial bound in estimating $S(\psi, f; \chi, g)$. Weil [12] established the following classical bound for nondegenerate sums with polynomial f and rational g ,

$$(1.3) \quad |S(\psi, f; \chi, g)| \leq (\deg(f) + \ell - 1) q^{1/2},$$

where ℓ is the number of distinct zeros and (non-infinite) poles of g in $\overline{\mathbb{F}_p}$. The constant $\deg(f) + \ell - 1$ is just the degree of the Artin L -function (a polynomial) associated with the sum when $p \nmid \deg(f)$. Weil used class-field theory and the Riemann Hypothesis for the zeta-function associated with an abelian extension of the function field $\mathbb{F}_q(t)$ in order to obtain (1.3). Elementary proofs were established later by Stepanov [8], [9], [10] with refinements by Schmidt [6]; the method is described in full in the monographs of Schmidt [7] and Lidl & Niederreiter [3].

Perel'muter [4] extended the result of Weil to the case of rational f obtaining the upper bound $(\deg(f)_\infty + \ell + \ell' - 2)\sqrt{q}$ where $(f)_\infty = \sum_{i=1}^{\ell'} n_i P_i$, the P_i being the poles of f (including ∞) of multiplicity n_i respectively, and $\deg(f)_\infty = \sum_{i=1}^{\ell'} n_i$. For polynomial f , $\deg(f) = \deg(f)_\infty$. The constant in his bound sometimes exceeds the degree of the associated L -function. For instance, for the generalized Kloosterman sum with $f(x) = \sum_{j=-N}^M b_j x^j$, $p \nmid NM$, and $g(x) = x$, the constant is $(M + N + 1)$, one more than the

Date: October 9, 2008.

1991 Mathematics Subject Classification. 11L07; 11L03.

Key words and phrases. exponential sums.

degree of the L -function; see Katz & Laumon [2]. Castro and Moreno [1] showed more generally a saving of one on the Perel'muter constant for each pole of f which is also a zero or pole of g and thus realized the estimate,

$$(1.4) \quad |S(\psi, f; \chi, g)| \leq (\deg(f)_\infty + \ell + \ell' - \ell'' - 2) q^{\frac{1}{2}},$$

where ℓ'' is the number of non-infinite poles of f which are zeros or poles of g . (Their results are actually stated for sums over projective curves and so appear slightly different.)

The proofs of Perel'muter and Castro & Moreno employ the sophistication of algebraic geometry and an appeal to the Riemann Hypothesis for an appropriate L -function. We show here that the much more elementary approach of Stepanov & Schmidt can be used to obtain inequality (1.4). The first step of our proof is to generalize the classical method used for polynomial f to show that the L -function is a polynomial, $L(z) = \prod_{i=1}^{\deg L(z)} (1 - \omega_i z)$. Then we express the leading term of the L -function in terms of Gauss sums of the type $S(\psi, x; \chi, x)$ which of course are well known to be of modulus $q^{1/2}$, to establish the correct modulus for $\prod_{i=1}^{\deg L} \omega_i$. Finally we use the Stepanov method to make a precise estimate of the number of points in $\mathbb{F}_{q^n}$ with $tr_n(f(x)) = b$ and $N_n(g(x)) = a$ (for fixed a, b), where tr_n and N_n are the trace and norm on $\mathbb{F}_{q^n}$, from which we are able to obtain $|\omega_i| \leq q^{1/2}$. Putting this together with the modulus of the product of the ω_i gives us an exact formula for the number of ω_i with $|\omega_i| = q^{1/2}$ and the number with $|\omega_i| = 1$. The accounting of the number of ω_i of each modulus seems to be absent from the literature. Indeed Schmidt [7] and Lidl & Niederreiter [3] only prove the upper bound $|\omega_i| \leq q^{\frac{1}{2}}$.

In our proof it is useful to distinguish the case where $\deg(f) > 0$ and so we write

$$(1.5) \quad f(x) = p(x) + \frac{r(x)}{q(x)}, \quad \deg(r) < \deg(q), \quad M = \deg(p) \geq 0,$$

for polynomials $p(x), q(x), r(x)$ in $\mathbb{F}_q[x]$, $(r(x), q(x)) = 1$. If $\deg(f) \geq 0$ then $M = \deg(f)$, while if $\deg(f) < 0$ then $M = 0$. It is also important to distinguish the poles of $f(x)$ that are also poles or zeros of $g(x)$ and so we factor $q(x)$ and $g(x)$ in $\mathbb{F}_q(x)$ as

$$(1.6) \quad q(x) = \prod_{i=1}^R f_i^{e_i}, \quad g(x) = \prod_{i=1}^R f_i^{w_i} \prod_{i=1}^r g_i^{s_i},$$

for distinct irreducible polynomials f_i, g_i and integers $e_i \geq 1$, $s_i \neq 0$ and w_i (possibly 0). We write

$$D_i = \deg(f_i), \quad d_i = \deg(g_i),$$

and

$$(1.7) \quad L = \sum_{i=1}^r d_i + \sum_{i=1}^R (e_i + 1) D_i.$$

When $f(x)$ is a polynomial we just have $g(x) = \prod_{i=1}^r g_i^{s_i}$, and

$$(1.8) \quad L = \sum_{i=1}^r d_i = \ell,$$

the number of distinct zeros and non-infinite poles of g . For the pure sums $\sum \psi(f(x))$ and $\sum \chi(g(x))$ we take $g \equiv 1$ and $f \equiv 0$ respectively. Obviously the remaining cases with f or g constant are either trivial or reduce to multiples of these pure sums. In this notation the Castro-Moreno bound (1.4) is

$$(1.9) \quad |S(\psi, f; \chi, g)| \leq (M + L - 1) q^{\frac{1}{2}}.$$

We obtain a slightly sharper upper bound in Theorems 1.1 and 1.2, taking into account the ω_i of modulus 1.

As it turns out, for the sums we are considering, a certain number of characteristic values ω_i can have modulus 1. To account for these values we define

$$(1.10) \quad U = U(g, \chi) = \sum_{\chi^{s_i} = \chi_0} d_i,$$

with $U = 0$ if $r = 0$ or $\chi^{s_i} \neq \chi_0$, the principal character, for all $i = 1, \dots, r$.

To state our result precisely we suppose that $g \equiv 1$ or non-constant with

$$(1.11) \quad g \neq ch^t, \text{ for any } t > 1, t|(q-1) \text{ and } h \in \mathbb{F}_q(x), c \in \mathbb{F}_q,$$

and that $f \equiv 0$ or non-constant with

$$(1.12) \quad M = 0 \text{ or } p \nmid M, \text{ and } p \nmid e_1 \cdots e_R \text{ if } R > 0.$$

Using the norm and trace from $\mathbb{F}_{q^n}$ to $\mathbb{F}_q$

$$N_n(x) = x^{1+q+\dots+q^{n-1}}, \quad \text{tr}_n(x) = x + x^q + \dots + x^{q^{n-1}},$$

to extend χ and ψ to characters of $\mathbb{F}_{q^n}$,

$$\chi_n(x) = \chi(N_n(x)), \quad \psi_n(x) = \psi(\text{tr}_n(x)),$$

it is natural to simultaneously consider the sums

$$S_n(\psi, f; \chi, g) = \sum_{x \in \mathbb{F}_{q^n} \setminus S} \chi_n(g) \psi_n(f), \quad n \geq 1.$$

The restriction (1.11) is quite natural since if $g = ch^t$ one would more properly write $S_n(\psi, f; \chi, g) = \chi(c)^n S_n(\psi, f; \chi^t, h)$. If restriction (1.12) does not hold then f can be straightforwardly reduced to a new rational function which is either constant or does satisfy (1.12) with a smaller value of $M + L - 1$. To see this, suppose that

$$b_\psi f(x) = h(x)^p - h(x) + b_\psi f_1(x)$$

for some h, f_1 in $\mathbb{F}_q(x)$, where (as above) $\psi(x) = e^{2\pi i \text{tr}_{\mathbb{F}_q/\mathbb{F}_p}(b_\psi x)}$. Then, since $\psi_n(x^p) = \psi_n(x)$ for any $x \in \mathbb{F}_{q^n}$ we have

$$S_n(\psi, f; \chi, g) = S_n(\psi, f_1; \chi, g),$$

less the value of $\chi_n(g)\psi_n(f_1)$ at any poles of f in $\mathbb{F}_{q^n}$ which are not poles of f_1 or g (we assume no new poles are introduced into f_1 by this process). Observing that elements in a finite field of characteristic p are automatically p -th powers, we can use this process to replace any monomial $cx^{p^j d}$ in $p(x)$ by the monomial $c'x^d$, $bc = (bc')^{p^j}$ and (by partial fractioning $\frac{r(x)}{q(x)}$ and replacing terms $\sum_{j=0}^{D_i-1} \frac{\sigma^j(A)}{(x-\sigma^j(\alpha_i))^{p^j d}}$ by $\sum_{j=0}^{D_i-1} \frac{\sigma^j(A')}{(x-\sigma^j(\alpha_i))^d}$ where α_i is a root of f_i and $\sigma : x \mapsto x^q$) reduce any multiplicities e_i in $r(x)/q(x)$ that are divisible by p . Clearly this process will not result in a constant unless

$$b_\psi f(x) = h(x)^p - h(x) + c$$

for some c in $\mathbb{F}_q$ and h in $\mathbb{F}_q(x)$. In particular, that will never happen if $p \nmid M$ or $p \nmid e_i$ for at least one of the $i = 1, \dots, R$. Thus conditions (1.11) and (1.12) assure us that the sum we are dealing with is nondegenerate.

We obtain the following for the mixed or pure additive sums:

Theorem 1.1. *Suppose that f, g are rational functions of the form (1.5) and (1.6). Suppose that f is non-constant and satisfies (1.12) and that $g \equiv 1$ or satisfies (1.11).*

Then there exist complex numbers $\omega_j = \omega_j(\psi, f; \chi, g)$, $1 \leq j \leq M + L - 1$ with $|\omega_j| = q^{1/2}$ or 1, such that for any $n \geq 1$,

$$(1.13) \quad S_n(\psi, f; \chi, g) = - \sum_{j=1}^{M+L-1} \omega_j(\psi, f; \chi, g)^n.$$

If $M > 0$, or $M = 0$ and $\deg(g) \neq 0$ with $\chi^{\deg(g)} \neq \chi_0$, then $|\omega_j| = q^{\frac{1}{2}}$ for $(M+L-1-U)$ of the ω_j , with $|\omega_j| = 1$ for the remaining U , where U is as defined in (1.10).

If $M = 0$ and $\deg(g) = 0$ or $\deg(g) \neq 0$ but $\chi^{\deg(g)} = \chi_0$, then $|\omega_j| = q^{\frac{1}{2}}$ for $(M+L-2-U)$ of the ω_j , with $|\omega_j| = 1$ for the remaining $(U+1)$.

If $\chi^{s_i} = \chi_0$, the principal character, for some of the $i = 1, \dots, r$ then it is of course more natural to consider the sum with those g_i simply omitted from g (this sum will have all its $|\omega_j| = q^{\frac{1}{2}}$ when $M > 0$) less a sum over the roots of the removed g_i that lie in $\mathbb{F}_{q^n}$, these terms accounting for the U additional values $|\omega_j| = 1$. The extra $|\omega_j| = 1$ that can occur when $M = 0$ and $\deg(g) = 0$ is illustrated by the example

$$\sum_{x \neq 0, 1} \chi_n \left(\frac{x-1}{x} \right) \psi_n \left(\frac{1}{x(x-1)} \right) = \psi_n(-2) \sum_{x \neq 0} \chi_n(x) \psi_n \left(x + \frac{1}{x} \right) - 1,$$

obtained using the substitution $x \rightarrow 1/(1-x)$. For the sum on the left, $M = U = 0$, $L = 4$, so there are three ω_j , two of modulus $q^{1/2}$ and one of modulus 1. For the sum on the right, $M = 1$, $U = 0$, $L = 2$, so there are two ω_j each of modulus $q^{1/2}$.

For the pure multiplicative sum, $f \equiv 0$, we obtain:

Theorem 1.2. *Suppose that g is a non-constant rational function of the form (1.6) of degree d and satisfying (1.11), and χ is a multiplicative character on $\mathbb{F}_q$ with $\chi \neq \chi_0$. Suppose that $L > U$. Then there exist $(L-1)$ non-zero complex numbers $\omega_j = \omega_j(\chi, g)$, $j = 1, \dots, L-1$, with $|\omega_j| = q^{\frac{1}{2}}$ or 1, such that for all $n \geq 1$*

$$(1.14) \quad \sum_{x \in \mathbb{F}_{q^n} \setminus \mathcal{S}} \chi_n(g(x)) = - \sum_{j=1}^{L-1} \omega_j(\chi, g)^n.$$

If $\chi^d \neq \chi_0$ then $|\omega_j| = q^{\frac{1}{2}}$ for $(L-1-U)$ of the ω_j with $|\omega_j| = 1$ for the remaining U .

If $\chi^d = \chi_0$ then $|\omega_j| = q^{\frac{1}{2}}$ for $(L-2-U)$ of the ω_j with $|\omega_j| = 1$ for the remaining $(U+1)$.

The need for the extra $|\omega_i| = 1$ when $\chi^d = \chi_0$ is illustrated by the example

$$(1.15) \quad \sum_{x \in \mathbb{F}_{q^n}} \chi_n(x^{e_1}(x-\alpha_2)^{e_2} \cdots (x-\alpha_L)^{e_L})$$

where the $\alpha_2, \dots, \alpha_L$ are in $\mathbb{F}_q^*$, the $e_i \geq 1$, and $\chi^{e_1+\dots+e_L} = \chi_0$, since the sum can be straightforwardly rewritten in the form

$$\sum_{x \in \mathbb{F}_{q^n}} \chi_n \left(\left(\frac{x-\alpha_2}{x} \right)^{e_2} \cdots \left(\frac{x-\alpha_L}{x} \right)^{e_L} \right) = -1 + \sum_{x \in \mathbb{F}_{q^n}} \chi_n((1-\alpha_2 x)^{e_2} \cdots (1-\alpha_L x)^{e_L}),$$

after the substitution $x \mapsto 1/x$.

2. USING THE ARTIN L-FUNCTION

We begin with a standard line of argument found for instance in [3, p195-197] for the case of polynomials.

Suppose that $f, g \in \mathbb{F}_q(x)$. Let $\mathcal{S}$ denote the poles of f and g , and

$$\mathcal{P} = \{F \in \mathbb{F}_q[x] : F \text{ monic, } F(\alpha) \neq 0 \text{ for } \alpha \text{ in } \mathcal{S}\},$$

$$\mathcal{P}_t = \{F \in \mathcal{P} : \deg(F) = t\}.$$

For a polynomial $F = \prod_{i=1}^t (x - \lambda_i) \in \mathcal{P}$ define

$$\lambda(\psi, f, \chi, g)(F) = \chi(g(\lambda_1) \cdots g(\lambda_t)) \psi(f(\lambda_1) + \cdots + f(\lambda_t)),$$

with $\lambda(1) = 1$, and

$$(2.1) \quad T_n = T_n(\psi, f, \chi, g) = \sum_{F \in \mathcal{P}_n} \lambda(F).$$

Since $\lambda(F)$ is totally multiplicative we have an Artin L-function

$$(2.2) \quad L(z) = \sum_{F \in \mathcal{P}} \lambda(F) z^{\deg(F)} = 1 + \sum_{j=1}^{\infty} T_j z^j = \prod_{\text{irred. } G \in \mathcal{P}} \left(1 - \lambda(G) z^{\deg(G)}\right)^{-1}.$$

If $T_j = 0$ for all $j > I$ then $L(z)$ is a polynomial

$$L(z) = \prod_{i=1}^I (1 - w_i z)$$

for some $w_1, \dots, w_I \in \mathbb{C}$ with $(-1)^I w_1 \cdots w_I = T_I$.

Hence

$$z \frac{L'(z)}{L(z)} = - \sum_{i=1}^I \frac{z w_i}{(1 - w_i z)} = \sum_{j=1}^{\infty} \left(\sum_{i=1}^I -w_i^j \right) z^j.$$

and

$$z \frac{L'(z)}{L(z)} = \sum_{\text{irred. } G \in \mathcal{P}} \deg(G) \frac{\lambda(G) z^{\deg(G)}}{1 - \lambda(G) z^{\deg(G)}} = \sum_{j=1}^{\infty} \left(\sum_{\substack{\text{irred. } G \in \mathcal{P} \\ \deg(G) | j}} \deg(G) \lambda(G)^{j/\deg(G)} \right) z^j.$$

Now if $\gamma \in \mathbb{F}_{q^n} \setminus \mathcal{S}$ has minimum polynomial G_γ over $\mathbb{F}_q$, then $\deg(G_\gamma) | n$ and

$$\chi_n(g(\gamma)) \psi_n(f(\gamma)) = \lambda(G_\gamma)^{n/\deg(G_\gamma)}.$$

Likewise for the $\deg(G_\gamma)$ conjugates of γ . Thus

$$(2.3) \quad S_n(\psi, f; \chi, g) = \sum_{\substack{\text{irred. } G \in \mathcal{P} \\ \deg(G) | n}} \deg(G) \lambda(G)^{n/\deg(G)} = - \sum_{i=1}^I w_i^n.$$

The following shows that $L(z)$ is typically a polynomial of degree $M + L - 1$:

Lemma 2.1. *Suppose that f, g are rational functions of the form (1.5) and (1.6). Suppose that $M > 0$ with $p \nmid M$, or $R > 0$ with $p \nmid e_i$ for some $i = 1, \dots, R$, or $r > 0$ with $\chi^{s_i} \neq \chi_0$ for some $i = 1, \dots, r$.*

Then $T_t(\psi, f, \chi, g) = 0$ for $t \geq M + L$.

If $M > 0$ and $p \nmid M e_1 \cdots e_R$ then $|T_{M+L-1}| = q^{\frac{1}{2}(M+L-1-U)}$.

Proof. Replacing g_i^{-1} by g_i^{q-2} etc. as necessary (and observing that $\chi(0) = 0$) we may plainly assume that $g(x)$ is a polynomial. The case $L = 0$ is just the classical case of a pure additive exponential sum with f being a polynomial and $g \equiv 1$ for which elementary proofs of the result are well known (see eg. Stepanov [11, ex. 1.3, #14]) and so we assume (at first) that $L > 0$. We will point out later, the simplification one makes for the case $L = 0$. Let $\alpha_i, i = 1, \dots, \sum_{i=1}^r d_i + \sum_{i=1}^R D_i$ denote the poles of f and any zeros of g . We write $\beta_{ij}, j = 1, \dots, d_i$ for the zeros of $g_i, i = 1, \dots, r$, ordered so that $\beta_{ij} = \sigma^{j-1}(\beta_{i1})$ where $\sigma(x) = x^q$, and $\gamma_{ij}, j = 1, \dots, D_i$ for the zeros of $f_i, i = 1, \dots, R$ with $\gamma_{ij} = \sigma^{j-1}(\gamma_{i1})$. Since g is now a polynomial, the set of poles of f and g is here just the set of poles of f and

$$\mathcal{S} = \{\gamma_{ij} : 1 \leq i \leq r, 1 \leq j \leq D_i\}.$$

Then $T_t = \sum \lambda(F)$ is a sum over polynomials F of the form

$$F(x) = \prod_{i=1}^t (x - \lambda_i) = x^t + u_1 x^{t-1} + \cdots + u_t,$$

with $u_1, \dots, u_t$ in $\mathbb{F}_q$ and $F(\gamma_{ij}) \neq 0$.

Recalling Waring's formulae [3, Theorem 1.76], and putting $u_i = 0$ for $i > t$,

$$\sum_{k=1}^t \lambda_k^j = \sum_{v_1+2v_2+\dots+jv_j=j} \frac{j(v_1+\dots+v_j-1)!}{v_1!\dots v_j!} (-u_1)^{v_1} \dots (-u_j)^{v_j}, \quad j > 0,$$

and writing $p(x) = \sum_{j=0}^M b_j x^j$, $b_M \neq 0$, we have

$$\sum_{i=1}^t p(\lambda_i) = -M b_M u_M + Q_0(u_1, \dots, u_{M-1}),$$

for some polynomial Q_0 . Decomposing into partial fractions

$$\frac{r(x)}{q(x)} = \sum_{i=1}^R \sum_{j=1}^{D_i} \sum_{l=1}^{e_i} \frac{\sigma^{j-1}(c_{il})}{(x - \gamma_{ij})^l},$$

for appropriate c_{il} in $\mathbb{F}_q^{D_i}$ with the $c_{ie_i} \neq 0$. We define an operator

$$(2.4) \quad D^r \left(\sum_{i=1}^j a_i x^i \right) = \sum_{i=r}^j \binom{i}{r} a_i x^{i-r}$$

on polynomials over $\mathbb{F}_q$ and note (taking successive derivatives of F) that for any r ,

$$\frac{D^r(F(x))}{F(x)} = \sigma_r \left(\frac{1}{x - \lambda_1}, \frac{1}{x - \lambda_2}, \dots, \frac{1}{x - \lambda_t} \right) = (-1)^r \sigma_r \left(\frac{1}{\lambda_1 - x}, \frac{1}{\lambda_2 - x}, \dots, \frac{1}{\lambda_t - x} \right),$$

where σ_r denotes the r -th elementary symmetric polynomial. Writing $D^r(\gamma_{ij}) = D^r(F(x))_{x=\gamma_{ij}}$ and using the Waring formulae we obtain, for some polynomials Q_i ,

$$\begin{aligned} \sum_{k=1}^t \frac{r(\lambda_k)}{q(\lambda_k)} &= \sum_{i=1}^R \sum_{j=1}^{D_i} \sum_{l=1}^{e_i} \sigma^{j-1}(c_{il}) \sum_{k=1}^t \frac{1}{(\lambda_k - \gamma_{ij})^l} \\ &= \sum_{i=1}^R \sum_{j=1}^{D_i} \sum_{l=1}^{e_i} \sigma^{j-1}(c_{il}) \sum_{v_1+2v_2+\dots+l v_l=l} \frac{l(v_1+\dots+v_l-1)!}{v_1!\dots v_l!} \left(-\frac{D^1(\gamma_{ij})}{F(\gamma_{ij})} \right)^{v_1} \dots \left(-\frac{D^l(\gamma_{ij})}{F(\gamma_{ij})} \right)^{v_l} \\ &= \sum_{i=1}^R \text{tr}_{D_i} \left(-e_i c_{ie_i} \frac{D^{e_i}(\gamma_{i1})}{F(\gamma_{i1})} + Q_i \left(\frac{D^1(\gamma_{i1})}{F(\gamma_{i1})}, \dots, \frac{D^{e_i-1}(\gamma_{i1})}{F(\gamma_{i1})} \right) \right). \end{aligned}$$

Also

$$(2.5) \quad g(\lambda_1) \dots g(\lambda_t) = \prod_{k=1}^t \prod_{i=1}^R \prod_{j=1}^{D_i} (\lambda_k - \gamma_{ij})^{w_i} \prod_{i=1}^r \prod_{j=1}^{d_i} (\lambda_k - \beta_{ij})^{s_i}$$

$$(2.6) \quad = (-1)^{t \deg(g)} \prod_{i=1}^R N_{D_i}(F(\gamma_{i1}))^{w_i} \prod_{i=1}^r N_{d_i}(F(\beta_{i1}))^{s_i},$$

and so

$$(2.7) \quad \chi(g(\lambda_1) \dots g(\lambda_t)) = \chi(-1)^{t \deg(g)} \prod_{i=1}^R \chi_{D_i}^{w_i}(F(\gamma_{i1})) \prod_{i=1}^r \chi_{d_i}^{s_i}(F(\beta_{i1})).$$

Setting

$$m_i = \begin{cases} 1, & \text{if } \alpha_i = \beta_{jk} \text{ for some } j, k, \\ (e_j + 1), & \text{if } \alpha_i = \gamma_{jk} \text{ for some } j, k, \end{cases}$$

and recalling the definition of L in (1.7) we define an $L \times L$ matrix $\mathcal{A}$ with rows

$$\binom{j}{k} \alpha_i^{j-k}, \quad j = 0, \dots, L-1$$

for $k = 0, \dots, m_i - 1$, $i = 1, \dots, \sum_{i=1}^r d_i + \sum_{i=1}^R D_i$. To be clear, if $m_i = 1$ then there is a single row $[1, \alpha_i, \alpha_i^2, \dots, \alpha_i^{L-1}]$ associated with α_i , while if $m_i > 1$ then there are $e_i + 1$ rows $[1, \alpha_i, \alpha_i^2, \dots, \alpha_i^{L-1}]$, $[0, 1, 2\alpha_i, 3\alpha_i^2, \dots]$, $[0, 0, 1, 3\alpha_i, 6\alpha_i^2, \dots]$, etc. corresponding to α_i . This matrix has a generalized Vandermonde determinant

$$\det(\mathcal{A}) = \prod_{i < j} (\alpha_i - \alpha_j)^{m_i m_j} \neq 0.$$

(The evaluation of this determinant can be found in Rausch [5], attributed there by Schinzel to C. Meray 1867). Thus when $t \geq L$ for a given $u_1, \dots, u_{t-L}$ in $\mathbb{F}_q$ and ξ_i in $\mathbb{F}_{q^{d_i}}$, $i = 1, \dots, r$ and ζ_{ij} in $\mathbb{F}_{q^{D_i}}$, $i = 1, \dots, R$, $j = 0, \dots, e_i$, there will be a solution $u_{t-L+1}, \dots, u_t$ in $\mathbb{F}_q$ to the linear system

$$(2.8) \quad \begin{aligned} F(\beta_{ij}) &= \sigma^{j-1}(\xi_i), & i = 1, \dots, r, \quad j = 1, \dots, d_i, \\ F(\gamma_{ij}) &= \sigma^{j-1}(\zeta_{i0}), & i = 1, \dots, R, \quad j = 1, \dots, D_i, \\ D^k(\gamma_{ij}) &= \sigma^{j-1}(\zeta_{ik}), & i = 1, \dots, R, \quad j = 1, \dots, D_i, \quad k = 1, \dots, e_i. \end{aligned}$$

The solution $u_{t-L+1}, \dots, u_t$ is in $\mathbb{F}_q$ since the values are invariant under the automorphism σ . Thus summing over the polynomials in $\mathcal{P}_t$ corresponds to summing over the $u_1, \dots, u_{t-L}$ in $\mathbb{F}_q$, the ξ_i in $\mathbb{F}_{q^{d_i}}$ for $i = 1, \dots, r$ and ζ_{ij} in $\mathbb{F}_{q^{D_i}}$ for $i = 1, \dots, R$, $j = 1, \dots, e_i$ and ζ_{i0} in $\mathbb{F}_{q^{D_i}}$ for $i = 1, \dots, R$.

When $t \geq M + L$ the $u_1, \dots, u_M$ will be distinct from the $u_{t-L+1}, \dots, u_t$ (and the remaining $t - L - M$ variables if $t > M + L$) and this linear transformation gives

$$T_t = \chi(-1)^{t \deg(g)} q^{t-L-M} U_1 U_2 \sum_{\zeta_{10} \in \mathbb{F}_{q^{D_1}}^*, \dots, \zeta_{R0} \in \mathbb{F}_{q^{D_R}}^*} \chi_{D_1}^{w_1}(\zeta_{10}) \cdots \chi_{D_R}^{w_R}(\zeta_{R0}) \left(\prod_{i=1}^R U_3(i) \right)$$

where

$$U_1 = \prod_{i=1}^r \sum_{\xi_i \in \mathbb{F}_{q^{d_i}}} \chi_{d_i}^{s_i}(\xi_i),$$

$$U_2 = \left(\sum_{u_M \in \mathbb{F}_q} \psi(-M b_M u_M) \right) \left(\sum_{u_1, \dots, u_{M-1} \in \mathbb{F}_q} \psi(Q_0(u_1, \dots, u_{M-1})) \right),$$

and

$$U_3(i) = \left(\sum_{\zeta_{ie_i} \in \mathbb{F}_{q^{D_i}}} \psi_{D_i} \left(-e_i c_{ie_i} \frac{\zeta_{ie_i}}{\zeta_{i0}} \right) \right) \left(\sum_{\zeta_{i1}, \dots, \zeta_{i(e_i-1)} \in \mathbb{F}_{q^{D_i}}} \psi_{D_i} \left(Q_i \left(\frac{\zeta_{i1}}{\zeta_{i0}}, \dots, \frac{\zeta_{i(e_i-1)}}{\zeta_{i0}} \right) \right) \right).$$

Recall for multiplicative and non-trivial additive characters χ_1 and ψ_1 on a finite field $\mathbb{F}_{q_1}$ that $\sum_{x \in \mathbb{F}_{q_1}} \chi_1(x) = 0$ unless $\chi_1 = \chi_0$ the principal character on $\mathbb{F}_{q_1}$, and $\sum_{x \in \mathbb{F}_{q_1}} \psi_1(Ax) = 0$ unless $A = 0$. Hence if $p \nmid M$ we have $U_2 = 0$. If $p \nmid e_i$ for some $i = 1, \dots, R$ we have $U_3(i) = 0$, and if $r > 0$ with $\chi^{s_i} \neq \chi_0$ for some $i = 1, \dots, r$ we have $U_1 = 0$. Thus, under the hypotheses of the lemma we have $T_t = 0$ for all $t \geq L + M$ as claimed.

Hence suppose now that $M > 0$ and $t = M + L - 1$ with $p \nmid M e_1 \cdots e_R$. When $L \geq 1$ we need to rewrite u_M in terms of the $u_1, \dots, u_{M-1}, \xi_i, \zeta_{ij}$. From Cramer's rule applied to the linear system (2.8), one sees that

$$u_M = \sum_{i=1}^R \sum_{j=0}^{e_i} \text{tr}_{D_i}(A_{ij} \zeta_{ij}) + \sum_{i=1}^r \text{tr}_{d_i}(B_i \xi_i) + C_0 + \sum_{i=1}^{M-1} C_i u_i$$

with the $A_{ij} \in \mathbb{F}_{q^{D_i}}$, $B_i \in \mathbb{F}_{q^{d_i}}$, $C_i \in \mathbb{F}_q$ with the Vandermonde determinants $A_{ie_i}, B_i \neq 0$. Hence

$$T_{M+L-1} = \chi(-1)^{t \deg(g)} \sum_{\zeta_{10} \in \mathbb{F}_{q^{D_1}}^*, \dots, \zeta_{R0} \in \mathbb{F}_{q^{D_R}}^*} \left(\prod_{i=1}^R \chi_{D_i}^{w_i}(\zeta_{i0}) \psi_{D_i}(-Mb_M A_{i0} \zeta_{i0}) \right) U_1 U_2 \prod_{i=1}^R (U_3(i) U_4(i)),$$

with

$$\begin{aligned} U_1 &= \prod_{i=1}^r \left(\sum_{\xi_i \in \mathbb{F}_{q^{d_i}}} \chi_{d_i}^{s_i}(\xi_i) \psi_{d_i}(-Mb_M B_i \xi_i) \right), \\ U_2 &= \sum_{u_1, \dots, u_{M-1} \in \mathbb{F}_q} \psi \left(Q_0(u_1, \dots, u_{M-1}) - Mb_M \left(C_0 + \sum_{i=1}^{M-1} C_i u_i \right) \right), \\ U_3(i) &= \sum_{\zeta_{ie_i} \in \mathbb{F}_{q^{D_i}}} \psi_{D_i} \left(-e_i c_{ie_i} \frac{\zeta_{ie_i}}{\zeta_{i0}} - Mb_M A_{ie_i} \zeta_{ie_i} \right), \end{aligned}$$

and

$$U_4(i) = \sum_{\zeta_{i1}, \dots, \zeta_{i(e_i-1)} \in \mathbb{F}_{q^{D_i}}} \psi_{D_i} \left(Q_i \left(\frac{\zeta_{i1}}{\zeta_{i0}}, \dots, \frac{\zeta_{i(e_i-1)}}{\zeta_{i0}} \right) - Mb_M \sum_{j=1}^{e_i-1} A_{ij} \zeta_{ij} \right).$$

If $L = 0$ (i.e. f is a polynomial and $g \equiv 1$) we have the same formula with only the term U_2 . Now $U_3(i) = q^{D_i}$ if $\zeta_{i0} = -e_i c_{ie_i} / Mb_M A_{ie_i}$ and zero otherwise. In particular, we can assume these values for the ζ_{i0} . Recall (see Lidl & Niederreiter [3, Theorem 5.11] or Schmidt [7, Theorem 3A, pg 47]) that when $A \neq 0$ the sum $\left| \sum_{x \in \mathbb{F}_{q^{d_i}}} \chi_{d_i}^{s_i}(x) \psi_{d_i}(Ax) \right| = 1$ if $\chi_{d_i}^{s_i} = \chi_0$ and $q^{\frac{1}{2}d_i}$ otherwise. Hence $|U_1| = q^{\frac{1}{2}(\sum_{i=1}^r d_i - U)}$.

From the Waring formulae it is not hard to see that for a $0 < k < \frac{1}{2}M$ the dependence on a u_{M-k} in $\psi \left(Q_0(u_1, \dots, u_{M-1}) - Mb_M (C_0 + \sum_{i=1}^{M-1} C_i u_i) \right)$ will take the form,

$$\psi(u_{M-k} (Mb_M u_k + Q_{0k}(u_1, \dots, u_{k-1}))).$$

Hence, for a given choice of $u_1, \dots, u_{k-1}$, summing this over $u_{M-k} \in \mathbb{F}_q$ gives q for the single value $u_k = -Q_{0k}(u_1, \dots, u_{k-1}) / Mb_M$ and zero otherwise. Hence if M is odd we gain a contribution $q^{(M-1)/2}$ from successively summing over the $u_{M-1}, \dots, u_{(M+1)/2}$ (and successively fixing the value $u_1, \dots, u_{(M-1)/2}$). When M is even we similarly obtain $q^{\frac{1}{2}M-1}$ from successively summing over the $u_{M-1}, \dots, u_{\frac{1}{2}M+1}$ (and successively fixing the value $u_1, \dots, u_{\frac{1}{2}M-1}$) leaving a sum

$$\sum_{u_{\frac{1}{2}M} \in \mathbb{F}_q} \psi \left(\frac{1}{2} Mb_M u_{\frac{1}{2}M}^2 + Q_{0\frac{1}{2}M}(u_1, \dots, u_{\frac{1}{2}M-1}) u_{\frac{1}{2}M} \right),$$

where recall $\left| \sum_{x \in \mathbb{F}_q} \psi(Ax^2 + Bx + C) \right| = q^{\frac{1}{2}}$ when $A \neq 0$ (see for example Lidl & Niederreiter [3, Theorem 5.33] or Schmidt [7, Theorem 3C, pg 48]). Hence $|U_2| = q^{\frac{1}{2}(M-1)}$. In the same manner the $|U_4(i)| = (q^{D_i})^{\frac{1}{2}(e_i-1)}$ and

$$|T_{M+L-1}| = q^{\frac{1}{2}(\sum_{i=1}^r d_i - U)} q^{\frac{1}{2}(M-1)} \prod_{i=1}^R q^{D_i} q^{\frac{1}{2}D_i(e_i-1)} = q^{\frac{1}{2}(M+L-1-U)}.$$

□

For the pure multiplicative sums the value of $|T_{L-1}|$ is evaluated slightly differently:

Lemma 2.2. Suppose that $g(x)$ is a monic polynomial which splits completely in $\mathbb{F}_q[x]$:

$$g(x) = \prod_{i=1}^L (x - \alpha_i)^{e_i}, \quad d = \sum_{i=1}^L e_i,$$

and that $\chi^{e_i} \neq \chi_0$ for $i = 1, \dots, L$, then

$$|T_{L-1}| = \begin{cases} q^{\frac{1}{2}(L-1)}, & \text{if } \chi^d \neq \chi_0, \\ q^{\frac{1}{2}(L-2)}, & \text{if } \chi^d = \chi_0. \end{cases}$$

Proof. From (2.7) we have

$$T_{L-1} = \chi(-1)^{d(L-1)} \sum_{u_1, \dots, u_{L-1} \in \mathbb{F}_q} \chi(F(\alpha_1)^{e_1} \cdots F(\alpha_L)^{e_L})$$

where

$$F(x) = x^{L-1} + u_1 x^{L-2} + \cdots + u_{L-1}.$$

Observing $\det(\alpha_i^j)_{i=1, \dots, L-1}^{j=0, 1, \dots, L-2} = \prod_{1 \leq i < j \leq L-1} (\alpha_i - \alpha_j) \neq 0$, we can make a linear change of variables $u_1, \dots, u_{L-1} \rightarrow w_1, \dots, w_{L-1}$, such that

$$F(\alpha_i) = w_i, \quad i = 1, \dots, L-1.$$

Writing

$$F(x) = \prod_{i=1}^{L-1} (x - \alpha_i) + \sum_{i=1}^{L-1} F(\alpha_i) \prod_{j=1, j \neq i}^{L-1} \left(\frac{x - \alpha_j}{\alpha_i - \alpha_j} \right),$$

it is clear that this results in

$$F(\alpha_L) = A_0 + A_1 w_1 + \cdots + A_{L-1} w_{L-1}$$

with

$$A_0 = \prod_{i=1}^{L-1} (\alpha_L - \alpha_i) \neq 0, \quad A_i = \prod_{j=1, j \neq i}^{L-1} \left(\frac{\alpha_L - \alpha_j}{\alpha_i - \alpha_j} \right) \neq 0, \quad i = 1, \dots, L-1.$$

Hence

$$\begin{aligned} T_{L-1} &= \chi(-1)^{d(L-1)} \sum_{\substack{w_1, \dots, w_{L-1} \in \mathbb{F}_q \\ w_L = A_0 + A_1 w_1 + \cdots + A_{L-1} w_{L-1}}} \chi(w_1)^{e_1} \cdots \chi(w_L)^{e_L} \\ &= \chi(-1)^{dL - e_L} \chi(A_0)^d \chi(A_1)^{-e_1} \cdots \chi(A_{L-1})^{-e_{L-1}} \sum_{\substack{w_1, \dots, w_L \in \mathbb{F}_q \\ 1 = w_1 + \cdots + w_L}} \chi(w_1)^{e_1} \cdots \chi(w_L)^{e_L}, \end{aligned}$$

on making the substitutions $w_L = A_0 u_L$, $-A_i w_i = A_0 u_i$, $1 \leq i \leq L-1$. The latter sum is just a Jacobi sum and so, by Theorem 5.21 of [3], can be written in terms of Gauss sums $G(\chi, \psi) = \sum_{x \in \mathbb{F}_q} \psi(x) \chi(x)$,

$$\sum_{\substack{w_1, \dots, w_L \in \mathbb{F}_q \\ 1 = w_1 + \cdots + w_L}} \chi(w_1)^{e_1} \cdots \chi(w_L)^{e_L} = \begin{cases} G(\chi^d, \psi)^{-1} \prod_{i=1}^L G(\chi^{e_i}, \psi), & \text{if } \chi^d \neq \chi_0, \\ -q^{-1} \prod_{i=1}^L G(\chi^{e_i}, \psi), & \text{if } \chi^d = \chi_0, \end{cases}$$

for any non-trivial additive character ψ . The result follows since $|G(\chi, \psi)| = q^{\frac{1}{2}}$ for non-trivial ψ and non-principal χ . $\square$

3. USING STEPANOV'S METHOD TO BOUND THE w_i

Suppose that f, g are non-constant rational functions satisfying (1.11) and (1.12) and $\mathcal{S}$ is the set of poles of f and g . For a in $\mathbb{F}_q^*$ and b in $\mathbb{F}_q$ define

$$s_n(f, b; g, a) = \{x \in \mathbb{F}_{q^n} \setminus \mathcal{S} : \text{tr}_n(f(x)) = b, N_n(g(x)) = a\}.$$

Observe that

$$\sum_{\text{all } \chi} \chi(x) = \begin{cases} q-1, & \text{if } x = 1, \\ 0, & \text{otherwise,} \end{cases} \quad \sum_{\text{all } \psi} \psi(x) = \begin{cases} q, & \text{if } x = 0, \\ 0, & \text{otherwise,} \end{cases}$$

and write ψ_0 for the trivial additive character. We sum $S_n(\psi, f; \chi, g)$ over all pairs (ψ, χ) satisfying the conditions of Theorem 1:

$$\begin{aligned} \sum_{\chi} \sum_{\psi \neq \psi_0} S_n(\psi, f; \chi, g) &= \sum_{\chi} \sum_{\psi} S_n(\psi, f; \chi, g) - \sum_{\chi} S_n(\psi_0, f; \chi, g) \\ &= q(q-1)s_n(f, 0; g, 1) - (q-1)\#\{x \in \mathbb{F}_{q^n} \setminus \mathcal{S} : N_n(g(x)) = 1\} \\ &= (q-1) \sum_{b \in \mathbb{F}_q} (s_n(f, 0; g, 1) - s_n(f, b; g, 1)). \end{aligned}$$

Plainly then, if

$$(3.1) \quad \left| s_n(f, b; g, a) - \frac{q^n}{q(q-1)} \right| < C(f, g, q) q^{n/2}$$

for each $a \in \mathbb{F}_q^*$, $b \in \mathbb{F}_q$, we have by (2.3) and Lemma 2.1

$$\left| \sum_{\psi \neq \psi_0, \chi} \sum_{i=1}^{M+L-1} \omega_i(\psi, f; \chi, g)^n \right| = \left| \sum_{\psi \neq \psi_0, \chi} S_n(\psi, f; \chi, g) \right| < 2(q-1)^2 C(f, g, q) q^{n/2}$$

for all sufficiently large n , and we can immediately deduce that all the $|\omega_j(\psi, f; \chi, g)| \leq q^{\frac{1}{2}}$ (since, using the box principle or otherwise, there will be a subsequence of n where the arguments of the $\omega_j(\psi, f; \chi, g)^n$ are all close to zero; see also Lidl & Niederreiter [3, Lemma 6.55] or Schmidt [7, Lemma 6A, pg 57]). For the pure additive sum, $g \equiv 1$, we merely note that

$$\sum_{x \in \mathbb{F}_{q^n} \setminus \mathcal{S}} \psi_n(f(x)) = S_n(\psi, f; \chi_0, x) + \begin{cases} 0, & \text{if } 0 \in \mathcal{S}, \\ \psi(f(0)), & \text{if } 0 \notin \mathcal{S}. \end{cases}$$

For the pure multiplicative sums,

$$S_n(\chi, g) = \sum_{x \in \mathbb{F}_{q^n} \setminus \mathcal{S}} \chi_n(g(x)),$$

with g satisfying (1.11), and $\chi \neq \chi_0$, we have

$$\begin{aligned} \sum_{\chi \neq \chi_0} S_n(\chi, g) &= \sum_{\chi} S_n(\chi, g) - S_n(\chi_0, g) \\ &= (q-1)\#\{x \in \mathbb{F}_{q^n} \setminus \mathcal{S} : N_n(g(x)) = 1\} - \#\{x \in \mathbb{F}_{q^n} \setminus \mathcal{S} : N_n(g(x)) \neq 0\} \\ &= \sum_{\beta \in \mathbb{F}_q^*} (\#\{x \in \mathbb{F}_{q^n} \setminus \mathcal{S} : N_n(g(x)) = 1\} - \#\{x \in \mathbb{F}_{q^n} \setminus \mathcal{S} : N_n(g(x)) = \beta\}) \\ &= \sum_{b \in \mathbb{F}_q} \sum_{\beta \in \mathbb{F}_q^*} (s_n(x, b; g, 1) - s_n(x, b; g, \beta)) \end{aligned}$$

and the result follows from the case above with $f = x$.

In order to prove (3.1) we shall need (1.11) and the following restriction on f :

$$(3.2) \quad y^q - y + f(x) \text{ is absolutely irreducible.}$$

Fortunately this will automatically hold for the non-constant reduced f we need to consider:

Lemma 3.1. *Let f be a rational function of type (1.5) with factorization of type (1.6). If $M > 0$ and $p \nmid M$, or $R > 0$ and $p \nmid e_i$ for some $i = 1, \dots, R$, then f satisfies (3.2).*

Proof. Write $f = f_1/f_2$ for coprime polynomials f_1, f_2 with $d_1 = \deg f_1, d_2 = \deg f_2$.

Suppose first that $M > 0$ with $p \nmid M = d_1 - d_2$. Rewrite $z^q - z + f = 0$ in the form $(zf_2(x))^q - f_2(x)^{q-1}(zf_2(x)) + f_1(x)f_2(x)^{q-1} = 0$. For polynomials g_i define

$$\kappa(z^d + g_1(x)z^{d-1} + \dots + g_d(x)) = \max_{1 \leq i \leq d} \frac{\deg(g_i)}{i}.$$

Then we have

$$\kappa(z^q - f_2(x)^{q-1}z + f_1(x)f_2(x)^{q-1}) = \max \left\{ \frac{(q-1)d_2}{q-1}, \frac{d_1 + (q-1)d_2}{q} \right\} = \frac{M + qd_2}{q}$$

with $(M + qd_2, q) = 1$ and absolute irreducibility follows from Schmidt [7, Theorem 1B, pg 92]. When $R > 0$ with $p \nmid e_i$ for some i then, by a change of variables $x \mapsto \frac{1}{x} + \alpha$ where α is a root of f_2 of multiplicity e_i , we can replace f by a rational function of degree e_i and apply the first case. $\square$

We use Stepanov's method to obtain (3.1):

Theorem 3.1. *Let $f = f_1/f_2, g = g_1/g_2$ in $\mathbb{F}_q(x)$ and set $d = \max\{d_1, d_2\}$ and $l = \max\{l_1, l_2\}$, where $d_i = \deg(f_i), l_i = \deg(g_i)$. Suppose that f, g satisfy (1.11) and (3.2). For any $n \geq 7, a \in \mathbb{F}_q^*$ and $b \in \mathbb{F}_q$*

$$(3.3) \quad \left| \#s_n(f, b; g, a) - \frac{q^n}{q(q-1)} \right| < 2(d+l)^2 q^{\frac{5}{2}} q^{n/2}.$$

We remark that having established Theorems 1.1 and 1.2, the upper bound in (3.3) can then be improved to $(M + L - 1)q^{n/2}$ for all $n \geq 1$. The condition (1.11) prevents cases such as $g(x) = x^l$ with $(l, q-1) > 1$ (and f a Laurent polynomial say) where $\sum_{b \in \mathbb{F}_q} s_n(f, b; g, a)$ will not be close to $q^n/(q-1)$ but equal $(l, q-1)(q^n-1)/(q-1)$ or zero. The condition (3.2) prevents cases such as $f = h^q - h$ where $\sum_{a \in \mathbb{F}_q^*} s_n(f, b; g, a)$ will not be close to q^{n-1} for each b but q^n for $b = 0$ and zero otherwise.

The idea of the Stepanov method is to construct a non-zero polynomial having a root of high multiplicity at each element of the set $s_n(f, b; g, a)$. We write $u = \lfloor n/2 \rfloor$ and $\rho_n = (q^n - 1)/(q - 1)$. Consider the polynomials

$$(3.4) \quad F(x) = \sum_{i=0}^{q-1} \sum_{j=0}^{q-2} \sum_{k=0}^K e_{ijk}(x) \Phi_i(x) \Lambda_j(x) x^{q^n k},$$

where the

$$\Phi_i(x) = f_2(x)^{q^{n-1}(q-1)} \left(\sum_{t=u}^{n-1} f(x)^{q^t} \right)^i, \quad \Lambda_j(x) = g_2(x)^{(q-2)\rho_n} g(x)^{\rho_n j},$$

the $e_{ijk}(x) = \sum_{t=0}^{S-1} a_{ijk t} x^t$ are polynomials in $\mathbb{F}_q[x]$, and

$$(3.5) \quad S = q^{n-2}, \quad K = \left\lfloor \frac{q^{u-2}}{d+l} \right\rfloor.$$

We can assume that

$$(3.6) \quad 2(d+l)^2 q^{\frac{n}{2} + \frac{5}{2}} < q^n,$$

otherwise the bound in (3.3) is trivial. In particular $K \geq 2(d+l) \geq 4$.

We show the existence of a set of $a_{ijk t}$ such that F is a non-zero polynomial with a root of order $R = q(q-1)K < q^u$ at each x in $s_n(f, b; g, a)$. The restrictions (1.11),

(3.2) are to ensure that F is not identically zero unless all the $a_{ijk} = 0$. It is perhaps worth noting that in the special case that $d_1 > d_2$ and $l_1 > l_2$ with $p \nmid (d_1 - d_2)$ and $(l_1 - l_2, q - 1) = 1$ this is straightforward since the terms constituting F will have different degrees (cf. condition (i) in Schmidt [7, Theorem 2G, pg 45]). Replacing the original $f(x)$, $g(x)$ by $f(x - c)$, $g(x - c)$ for some c in $\mathbb{F}_{q^n}$ as necessary, we shall suppose that $f(0)$, $g(0)$ are defined and non zero (since $q^n > 2d + 2l$ there will certainly be such a c).

Lemma 3.2. *Suppose that f, g are rational functions over $\mathbb{F}_q$ satisfying (1.11), (3.2) and (3.6), and that $f(0), g(0)$ are defined and nonzero. Suppose further that*

$$F(Z, Y) = \sum_{i=0}^{q-1} \sum_{j=0}^{q-2} \left(\sum_{k=0}^K e_{ijk}(x) x^{q^n k} \right) Z^i Y^j$$

where the $e_{ijk}(x)$ in $\overline{\mathbb{F}}_q[x]$ have $\deg(e_{ijk}) < q^{n-2}$. If $Z_1 = \sum_{t=u}^{n-1} f(x)^{q^t}$ and $Y_1 = g(x)^{q^n}$ then $F(Z_1, Y_1) \neq 0$ unless the $e_{ijk}(x) \equiv 0$ for all i, j, k .

Proof. Let $Y_1, Y_2, \dots, Y_{q-1}$ and $Z_1, \dots, Z_q$ denote the roots of $Y^{q-1} = g(x)^{q^n-1}$ and $Z^q - Z = f(x)^{q^n} - f(x)^{q^u}$ respectively,

$$\{Y_1, \dots, Y_{q-1}\} = \{\gamma g(x)^{q^n} : \gamma \in \mathbb{F}_q^*\},$$

$$\{Z_1, \dots, Z_q\} = \{f^{q^{n-1}} + f^{q^{n-2}} + \dots + f^{q^u} + \beta : \beta \in \mathbb{F}_q\}.$$

Then $F(Z_1, Y_1) = 0$ gives

$$0 = \prod_{i=1}^q \prod_{j=1}^{q-1} F(Z_i, Y_j)$$

where this expression will be symmetric in the Z_i and symmetric in the Y_j (of degree at most $(q-1)^2$ in each Z_i and degree at most $q(q-2)$ in each Y_j). Thus the $Z_1, \dots, Z_q$ dependence will be polynomial of degree at most $(q-1)^2$ in the elementary symmetric polynomials

$$(-1)^q \sigma_q(Z_1, \dots, Z_q) = f(x)^{q^n} - f(x)^{q^u}, \quad (-1)^{(q-1)} \sigma_{q-1}(Z_1, \dots, Z_q) = -1,$$

and $(-1)^j \sigma_j(Z_1, \dots, Z_q) = 0, 1 \leq j \leq q-2$, and the $Y_1, \dots, Y_{q-1}$ dependence polynomial of degree at most $q(q-2)$ in the

$$(-1)^{(q-1)} \sigma_{q-1}(Y_1, \dots, Y_{q-1}) = -g(x)^{q^n-1}, \quad (-1)^j \sigma_j(Y_1, \dots, Y_{q-1}) = 0, 1 \leq j \leq q-2.$$

Hence we can write

$$(3.7) \quad 0 = \prod_{i=1}^q \prod_{j=1}^{q-1} F(Z_i, Y_j) = \sum_{i=0}^{(q-1)^2} \sum_{j=0}^{q(q-2)} \left(\sum_{k=0}^{K'} A_{ijk}(x) x^{q^n k} \right) \left(f(x)^{q^n} - f(x)^{q^u} \right)^i \left(g(x)^{q^n-1} \right)^j$$

for some polynomials A_{ijk} with $\deg(A_{ijk}) \leq q(q-1)q^{n-2}$. We note that the $A_{ij0}(x)$ are just the terms obtained from the $e_{ijk}(x)$ and so defining

$$\tilde{F}(Z, Y) = \sum_{j=0}^{q-2} \left(\sum_{i=0}^{q-1} e_{ij0}(x) Z^i \right) Y^j$$

we have

$$(3.8) \quad \prod_{i=1}^q \prod_{j=1}^{q-1} \tilde{F}(Z_i, Y_j) = \sum_{i=0}^{(q-1)^2} \sum_{j=0}^{q(q-2)} A_{ij0}(x) \left(f(x)^{q^n} - f(x)^{q^u} \right)^i \left(g(x)^{q^n-1} \right)^j.$$

Now, we can write $f(x)^{q^n} = f(0) + x^{q^n} \frac{h_1(x)}{f_2(x)^{q^n}}$ and $g(x)^{q^n-1} = \frac{g(0)}{g(x)} + x^{q^n} \frac{h_2(x)}{g(x)g_2(x)^{q^n}}$ for some polynomials $h_1(x), h_2(x)$. Substituting these expressions into (3.7) we obtain that

x^{q^n} divides the polynomial

$$P(x) = f_2(x)^{q^u(q-1)^2} g_1(x)^{q(q-2)} \sum_{i=0}^{(q-1)^2} \sum_{j=0}^{q(q-2)} A_{ij0}(x) \left(\frac{g(0)}{g(x)} \right)^j \left(f(0) - f(x)^{q^u} \right)^i.$$

But from (3.6) we have $\deg(P) \leq (q-1)q^{n-1} + lq(q-2) + dq^u(q-1)^2 < q^n$ and so $P(x) \equiv 0$. Then, from (3.8), we have

$$0 \equiv P(x) = f_2(x)^{q^u(q-1)^2} g_1(x)^{q(q-2)} \prod_{i=1}^q \prod_{j=1}^{q-1} \tilde{F}(z_i, y_j)$$

where $y_1, \dots, y_{q-1}$ are the roots of $y^{q-1} = g(0)/g(x)$ and $z_1, \dots, z_q$ are the roots of $z^q - z = (f(0) - f(x)^{q^u})$. Therefore $\tilde{F}(z_I, y_J) = 0$ for some I, J , that is,

$$\sum_{j=0}^{q-2} \left(\sum_{i=0}^{q-1} e_{ij0}(x) z_I^i \right) y_J^j \equiv 0$$

for some I, J . Now condition (1.11) ensures that

$$(3.9) \quad [\overline{\mathbb{F}}_q(x, y_J) : \overline{\mathbb{F}}_q(x)] = q-1,$$

since, see for example Lemma 6.54 of Lidl & Niederreiter, when $G(x)$ is a polynomial, $y^{q-1} - G(x)$ is absolutely irreducible unless $G(x) = h(x)^t$ for some $h(x) \in \overline{\mathbb{F}}_q[x]$ and $t > 1$ with $t|(q-1)$, and y_J satisfies $(g_1(x)y)^{q-1} - g(0)g_2(x)g_1(x)^{(q-1)-1}$. Condition (3.2) similarly ensures that

$$(3.10) \quad [\overline{\mathbb{F}}_q(x, z_I) : \overline{\mathbb{F}}_q(x)] = q,$$

since, writing $b - b^q = f(0)$ for some b in $\overline{\mathbb{F}}_q$,

$$z^q - z + f(x)^{q^u} - f(0) = (z + f(x)^{q^{u-1}} + \dots + f(x) + b)^q - (z + f(x)^{q^{u-1}} + \dots + f(x) + b) + f(x).$$

Hence

$$[\overline{\mathbb{F}}_q(x, z_I, y_J) : \overline{\mathbb{F}}_q(x)] = q(q-1),$$

(it is clearly $\leq q(q-1)$ but must be divisible by the degrees (3.9) and (3.10)). In particular

$$[\overline{\mathbb{F}}_q(x, z_I, y_J) : \overline{\mathbb{F}}_q(x, z_I)] = q-1,$$

and the $\sum_{i=0}^{q-1} e_{ij0}(x) z_I^i \equiv 0$ for $j = 0, \dots, q-2$. Hence from (3.10) we must have $e_{ij0}(x) \equiv 0$ for all i, j . Dividing by x^{q^n} and repeating plainly gives $e_{ijk}(x) \equiv 0$ for all i, j, k . $\square$

Proof of Theorem 3.1. Letting D^r again be the operator $D^r \left(\sum_{j=0}^N a_j x^j \right) = \sum_{j=r}^N a_j \binom{j}{r} x^{j-r}$ we observe that any polynomial F has a root of order at least R at γ iff $D^r F(\gamma) = 0$ for $r = 0, \dots, R-1$. For $0 \leq r < q^u$ it can be readily verified (see Corollary 6.50 of [3]) that $D^r(p(x, x^{q^u})) = D^r(p(x, y))|_{y=x^{q^u}}$, where on the right-hand side the operator is defined as above with the a_j regarded as polynomials in y . Thus, in order for a polynomial F of type (3.4) to have a zero of order R at each point of $s_n(f, b; g, a)$ it is enough to ensure that

$$g_2^{-(q-2)\rho_n} f_2^{-(q^{n-1}-q^{u-1})(q-1)} (g_1 g_2)^r \sum_{i=0}^{q-1} \sum_{j=0}^{q-2} \sum_{k=0}^K D^r(e_{ijk}(x) \Lambda_j(x)) \Phi_i(x) x^{q^n k} = 0,$$

for all $r = 0, \dots, R-1$, and each x in $s_n(f, b; g, a)$. Since for x in $s_n(f, b; g, a)$ we have $g(x)^{\rho_n j} x^{q^n k} = a^j x^k$ the $g_2(x)^{-(q-2)\rho_n} (g_1 g_2)^r D^r(e_{ijk}(x) \Lambda_j(x)) x^{q^n k}$ reduce to polynomials of degree less than $S + (l_1 + l_2)r + K$ in x , while the

$$f_2(x)^{-(q^{n-1}-q^{u-1})(q-1)} \Phi_i(x) = f_2(x)^{q^{u-1}(q-1)} \left(b - \sum_{t=1}^{u-1} f(x)^{q^t} \right)^i$$

give polynomials of degree $\max\{0, (d_1 - d_2)q^{u-1}i\} + d_2(q-1)q^{u-1} \leq d(q-1)q^{u-1}$.

Our strategy then is to make the resulting reduced polynomials of degree at most $(S-1+d(q-1)q^{u-1}+r(l_1+l_2)+K)$ vanish identically for $r=0, 1, \dots, R-1$. Now the coefficients of the reduced polynomials are linear expressions in the a_{ijkt} (the coefficients of the $e_{ijk}(x)$), $0 \leq i \leq q-1$, $0 \leq j \leq q-2$, $0 \leq k \leq K$, $0 \leq t \leq S-1$. Hence we can make a root of order R at each x in $s_n(f, b; g, a)$ by making at most

$$R \left(S + d(q-1)q^{u-1} + \frac{1}{2}R(l_1+l_2) + K \right) < R(S + (d+l)q^u) \leq RS + (q-1)q^{2u-1}$$

homogeneous linear equations in the a_{ijkt} vanish. Since the number of variables a_{ijkt} ,

$$q(q-1)S(K+1) = RS + (q-1)q^{n-1},$$

exceeds this we are guaranteed the existence of a non-zero (by Lemma 3.2) polynomial F with the desired property. Hence, from degree considerations,

(3.11)

$$\begin{aligned} \#s_n(f, b; g, a) &\leq \frac{\deg(F)}{R} \leq \frac{q^n K + d(q-1)q^{n-1} + l(q-2)\rho_n + S}{R} \\ (3.12) \quad &\leq \frac{q^n K + (d+l)(q-1)q^{n-1}}{q(q-1)K} \leq \frac{q^n}{q(q-1)} + 2(d+l)^2 q^{n-u}. \end{aligned}$$

Observing that this holds for all a, b , and that the set $\mathcal{S}'$ of x in $\mathcal{S}$ or with $N_n(g(x)) = 0$ satisfies $|\mathcal{S}'| \leq d_2 + l_2 + l_1 < (d+l)^2$ we also have the lower bound

$$(3.13) \quad \#s_n(f, b; g, a) = q^n - |\mathcal{S}'| - \sum_{\substack{a', b' \in \mathbb{F}_q, a' \neq 0, \\ (a', b') \neq (a, b)}} \#s_n(f, b'; g, a')$$

$$(3.14) \quad \geq q^n - |\mathcal{S}'| - (q(q-1) - 1) \left(\frac{q^n}{q(q-1)} + 2(d+l)^2 q^{n-u} \right)$$

$$(3.15) \quad \geq \frac{q^n}{q(q-1)} - 2(d+l)^2 q^{n-u+2},$$

and the claim is plain. $\square$

Completion of the proof of Theorems 1.1 and 1.2. The expressions (1.13) and (1.14) follow at once from (2.3) and Lemma 2.1, and the bound $|\omega_j| \leq q^{\frac{1}{2}}$ from Theorem 3.1 as explained at the beginning of Section 3 (by Lemma 3.1 a non-constant f satisfying (1.12) will satisfy condition (3.2)).

It remains to determine the magnitudes of the ω_j . Let $g(x) = \prod_{j=1}^R f_i(x)^{w_i} \prod_{i=1}^r g_i(x)^{s_i}$ as in (1.6) with $d_i = \deg(g_i)$, $1 \leq i \leq r$, and suppose that $\chi^{s_i} = \chi_0$ for $i = 1, \dots, t$. Set $U = \sum_{i=1}^t d_i$. Then, writing $g(x) = G(x) \prod_{i=1}^t g_i(x)^{s_i}$, we have

$$S_n(\psi, f; \chi, g) = S_n(\psi, f; \chi, G) - \sum_{\alpha \in \mathcal{T}_n} \chi_n(G(\alpha)) \psi_n(f(\alpha))$$

where $\mathcal{T}_n$ is the set of zeros of the $g_i(x)$, $1 \leq i \leq t$, contained in $\mathbb{F}_{q^n}$. (By definition, the zeros of the $g_i(x)$ are never poles of f .) Now the zeros of $g_i(x)$ are in $\mathbb{F}_{q^n}$ if and only if $d_i | n$ and thus letting α_i denote a root of g_i we obtain,

$$S_n(\psi, f; \chi, g) = S_n(\psi, f; \chi, G) - \sum_{1 \leq i \leq t, d_i | n} d_i (\chi_{d_i}(G(\alpha_i)) \psi_{d_i}(f(\alpha_i)))^{n/d_i},$$

where the first sum $S_n(\psi, f; \chi, G) = -\sum_{j=1}^{M+L-U-1} \omega_j^n$ has none of its $\chi^{s_i} = \chi_0$ and the second sum can be written as a sum of U terms ω_j^n with $|\omega_j| = 1$ (of the form $\omega_j = e^{2\pi i l/d_i} (\chi_{d_i}(g(\alpha_i)) \psi_{d_i}(f(\alpha_i)))^{1/d_i}$, $l = 1, \dots, d_i$, $i = 1, \dots, t$). So it is enough to show the result when $U = 0$.

Suppose that $M > 0$. When $r = 0$ or none of the $\chi^{s_i} = \chi_0$ then from Lemma 2.1 we have $|\omega_j| \leq q^{\frac{1}{2}}$ with $\prod_{j=1}^{M+L-1} |\omega_j| = q^{\frac{1}{2}(M+L-1)}$ and so all the $|\omega_j| = q^{\frac{1}{2}}$.

Consider now the case where $M = 0$, that is, $\deg(f) \leq 0$. Observe that if we just want the magnitudes of the ω_j then we can always work in a fixed finite extension $\mathbb{F}_{q^t}$ of $\mathbb{F}_q$, regarding the $\mathbb{F}_q$ sums $S_{nt}(\psi, f; \chi, g)$ as the $\mathbb{F}_{q^t}$ sums $S_n(\psi_t, f; \chi_t, g)$ since this simply replaces the ω_j by ω_j^t (as can be seen by considering the corresponding $L(z)$). In particular, we may assume that f splits into linear factors over $\mathbb{F}_q$.

Suppose first that f is nonconstant, and therefore has a pole $\alpha \in \mathbb{F}_q$. Make the substitution $x \mapsto \frac{1}{x} + \alpha$ to obtain new f and g over $\mathbb{F}_q$, where the new f has $M = e_1 > 0$ and one less pole of multiplicity e_1 . When $\deg(g) \neq 0$ the factor $x^{\deg(g)}$ in the new g increases r to $(r+1)$, and $M+L-1$ is unchanged, and we obtain a sum equal to $-\sum_{j=1}^{M+L-1} \omega_j^n$ with none of the $|\omega_j| = 1$ if $\chi^{\deg(g)} \neq \chi_0$ and one of modulus 1 if $\chi^{\deg(g)} = \chi_0$, and the remaining $|\omega_j| = q^{\frac{1}{2}}$. If $\deg(g) = 0$ then r is unchanged and we obtain a sum equal to $-\sum_{j=1}^{M+L-2} \omega_j^n$ with $|\omega_j| = q^{\frac{1}{2}}$, but we also need to subtract the value at $x = 0$ from the new sum, giving the additional $|\omega_j| = 1$.

For the pure multiplicative case ($f \equiv 0$), Theorem 1.2, we assume again that $g(x)$ splits completely in $\mathbb{F}_q$. If $\chi^d \neq \chi_0$ then from Lemma 2.2 we have $\prod_{i=1}^{L-1} |\omega_i| = q^{\frac{1}{2}(L-1)}$ and the bound $|\omega_i| \leq q^{\frac{1}{2}}$ gives all $|\omega_i| = q^{\frac{1}{2}}$. If $\chi^d = \chi_0$ then transforming $x \mapsto (x + \alpha_1)$ reduces to a sum (1.15) and, as shown there, we gain an extra $\omega_i = -1$ and a sum with all $(L-2)$ of its $|\omega_i| = q^{\frac{1}{2}}$.

□

REFERENCES

- [1] F. Castro & C. Moreno, *Mixed exponential sums over finite fields*, Proc. Amer. Math. Soc. 128 (2000), 2529-2537.
- [2] N. M. Katz & G. Laumon, *Transformation de Fourier et majoration de sommes exponentielles*, I.H.E.S. Pub. Mathematiques 62 (1985), 145-202.
- [3] R. Lidl & H. Niederreiter, *Finite Fields*, Encyclopedia of mathematics and its applications, Addison-Wesley 1983.
- [4] G. I. Perel'muter, *Estimate of a sum along an algebraic curve*, Mat. Zametki 5 (1969), 373-380.
- [5] U. Rausch, *On a theorem of Dobrowolski about the product of conjugate numbers*, Colloq. Math. 50 (1985), 137-142.
- [6] W.M. Schmidt, *Zur Methode von Stepanov*, Acta Arith. 24 (1973), 347-367.
- [7] ———, *Equations over finite fields with applications, an elementary approach*, lecture notes in mathematics vol 536, Springer-Verlag 1976.
- [8] S. A. Stepanov, *The number of points of a hyperelliptic curve over a prime field (Russian)*, Izv. Akad. Nauk SSSR Ser. Mat. 33 (1969), 1171-1181.
- [9] ———, *Elementary method in the theory of congruences for prime modulus*, Acta Arith. 17 (1970), 231-247.
- [10] ———, *Estimates of rational trigonometric sums with prime denominators (Russian)*, Trudy Akad. Nauk 62 (1971), 346-371.
- [11] ———, *Arithmetic of Algebraic Curves*, Monographs in Contemporary Mathematics, Consultants Bureau, New York, 1994.
- [12] A. Weil, *On some exponential sums*, Proc. Nat. Acad. Sci. 34 (1948), 204-207.

DEPARTMENT OF MATHEMATICS, KANSAS STATE UNIVERSITY, MANHATTAN, KS 66506
E-mail address: cochrane@math.ksu.edu

DEPARTMENT OF MATHEMATICS, KANSAS STATE UNIVERSITY, MANHATTAN, KS 66506
E-mail address: pinner@math.ksu.edu